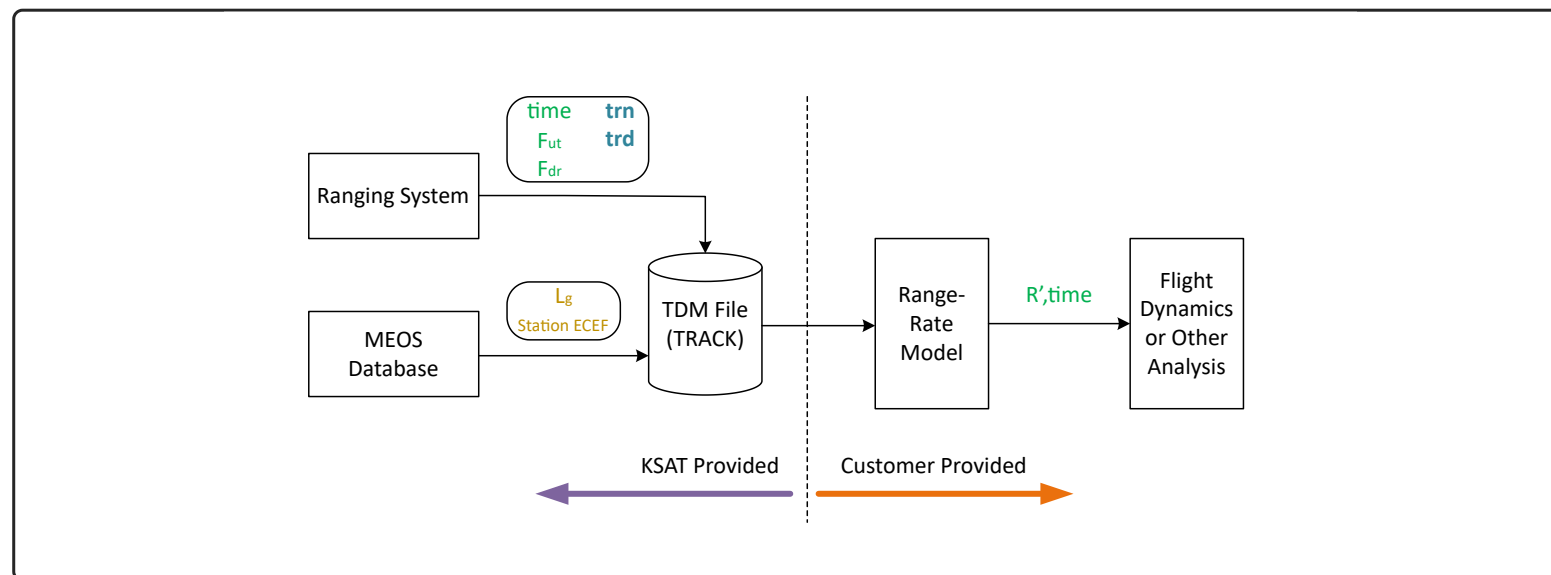


# Range Rate Model

The range rate model is used to convert carrier frequency measurements made by the baseband equipment into range rate measurements. Range rate is observable on the carrier as a Doppler shift. Complicating the generation of range rate is the fact that the uplink carrier is a different frequency than the downlink carrier, and thus the Doppler shift on the uplink is different than the downlink.

To support the range rate algorithm, the spacecraft transponder is presumed to perform a coherent translation of the uplink carrier to the downlink carrier using an integer turnaround ratio, so that Doppler shift on the uplink is preserved and measurable on the downlink.

The data flow of the range rate model is shown in the following figure. Models for troposphere and ionosphere are not included, since their rate of change is presumably low enough that their effect on range rate measurements is minimal.



KSAT provides raw, timestamped frequency measurements from the modem, as well as the turnaround ratio values. In addition, certain constants pertaining to the remote site are provided in the TDM file.

The uplink and downlink frequencies are tied to each other through a ratio of two integers known as the turnaround ratio. Multiple turnaround ratios are available, with the most common defined by [CCSDS 401.0-B-32](#). This specification includes ratios for cross-band operations, which allow, for example, coherent turnaround between uplink on S-band and downlink on X-band. However, these cross-band ratios are not currently supported by KSAT.

The available turnaround ratios are listed in the following table:

| BAND    | UPLINK RANGE    | DOWNLINK RANGE | TURNAROUND RATIO |
|---------|-----------------|----------------|------------------|
| S-band  | 2025-2110 MHz   | 2200-2300 MHz  | 240/221          |
| X-band  | 7145-7235 MHz   | 8400-8500 MHz  | 880/749          |
| Ka-band | 22.55-23.15 GHz | 25.5-27.0 GHz  | 2720/2407        |
| Ka-band | 22.55-23.15 GHz | 25.5-27.0 GHz  | 2760/2407        |
| Ka-band | 22.55-23.15 GHz | 25.5-27.0 GHz  | 2816/2407        |

When choosing carrier frequencies, it is recommended that the uplink frequency be an integer multiple of the turnaround ratio denominator, and the downlink frequency will then be an integer multiple of the numerator. This will reduce any error associated with tuning step size on the baseband modem.

The customer must implement the range rate model described [on this page](#) to produce the time-tagged range rate values prior to processing by flight dynamics or other required analysis.

A detailed derivation of the range rate model is provided [here](#)